



*Innovation & Entrepreneurship Hub for Educated Rural Youth **SURE ProEd** (formerly **SURE Trust**)*

SmartStock AI

Data-Driven Inventory Replenishment & Stock Optimization

The domain of the Project:

DATA ANALYTICS

GROUP -1

Team Mentors (and their designation):

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Senior Product Engineer at EsperGroup

&

Mr. Bhargavesh Dakka

GenAI Consultant at Deloitte USI

Submitted By

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BCA in Data Science and Artificial Intelligence

Babu Banarasi Das University

Period of the Project

February 2026 to September 2026



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Declaration

The project titled **SmartStock AI : Data-Driven Inventory Replenishment & Stock Optimization** has been mentored by **Ms. Nithyasri kannathal EL** ma'am and **Mr. Bhargavesh Dakka** sir, organised by SURE Trust, from February 2026 to September 2026 for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Submitted By:

Deepak Kumar

Mentor's Name: Ms. Nithyasri kannathal EL

Designation - Senior Product Engineer at EsperGroup

Mentor's Name: Mr. Bhargavesh Dakka

Designation - GenAI Consultant at Deloitte

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Executive Summary

SmartStock AI is a data-driven inventory decision-support system developed as the final project of a six-month Data Analytics Internship and Capstone Program with Sure Trust. The project focuses on using historical retail data to understand demand patterns, identify inventory risk, and provide restocking suggestions to support better inventory decisions.

The project uses the FreshRetailNet-50K dataset as its primary data source. The dataset contains historical information related to stores, products, sales, stock status, discounts, holidays, and contextual factors. A supporting Instacart Market Basket Analysis dataset was used separately to study purchase and reorder behaviour. The two datasets represent different retail environments and were not directly joined.

The project follows an end-to-end analytics workflow: **data preparation, exploratory data analysis, feature engineering, demand forecasting, inventory risk analysis, restocking recommendation, business intelligence, and application development**. A Random Forest Regressor was used to predict next-day demand for store-product combinations. The model achieved an MAE of 0.3889, RMSE of 0.6451, R^2 of 0.8814, and MAPE of 53.90%.

Based on the forecast and historical stockout patterns, the system categorizes inventory risk into **Low, Medium, and High** levels. It also provides a suggested restocking quantity using a simple seven-day cover rule:

Suggested Restocking Quantity = Predicted Daily Demand $\times$ 7

The project uses PostgreSQL for data storage and management, Power BI for business dashboards, and Streamlit for the interactive application. The application contains eight main pages covering the inventory dashboard, products, stores, demand forecasting, inventory risk, restocking, inventory data entry, and the AI Inventory Assistant.

The **Gemini API** is used as an explanation layer within the AI Inventory Assistant. It helps explain existing forecast, risk, and restocking results in simple language. It does not perform the main demand forecasting, which is handled by the Random Forest model.

SmartStock AI is designed as a **decision-support system rather than an autonomous purchasing system**. The manager reviews the available forecast, risk level, and restocking suggestion before making the final inventory decision. The deployed application uses Streamlit Community Cloud with Neon PostgreSQL, while project source code is maintained through GitHub.



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Introduction

2.1 Background and Context

Inventory management is an important part of retail operations. Businesses need to maintain enough stock to meet customer demand while avoiding unnecessary inventory. Changes in demand, stockouts, seasonal patterns, discounts, holidays, and other factors can make inventory planning difficult.

SmartStock AI was developed to apply data analytics and machine learning to this problem. The system uses historical retail data to understand sales and demand patterns, forecast next-day demand, identify inventory risk, and provide restocking suggestions.

The project was developed as the final capstone project of a six-month Data Analytics Internship and Capstone Program with Sure Trust. It brings together skills developed during the internship, including Python, SQL, data analysis, data visualization, machine learning, business intelligence, application development, and deployment.

The overall approach of the project can be represented as:

Data → Analysis → Forecast → Risk → Restocking → Decision

2.2 Problem Statement

Retail managers need to make inventory decisions based on changing demand and stock conditions. Relying only on historical sales figures or manual observation can make it difficult to identify products that may require attention.

The problem addressed by SmartStock AI is to develop a system that can use historical retail data to:

- Understand demand and sales patterns.
- Identify patterns related to stockouts.
- Forecast next-day demand.
- Identify products with different levels of inventory risk.
- Provide suggested restocking quantities.
- Present the results through dashboards and an interactive application.

The system is intended to support managers in making informed inventory decisions. It does not automatically place purchase orders or replace the manager's final decision.



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2.3 Project Scope and Limitations

The scope of SmartStock AI includes historical retail data analysis, demand forecasting, inventory risk analysis, restocking recommendations, business intelligence dashboards, and an interactive web application.

The project uses FreshRetailNet-50K as the primary dataset and Instacart Market Basket Analysis as a supporting dataset for purchase and reorder behaviour. These datasets represent different environments and are not directly joined.

The project has the following limitations:

- It uses historical retail data rather than live ERP inventory data.
- It does not provide real-time warehouse inventory information.
- Restocking suggestions use a simple seven-day cover rule.
- The system provides decision support rather than automatic purchase orders.
- The machine learning model is trained using the available historical data.
- New inventory information entered through the application is stored in PostgreSQL but does not automatically retrain the machine learning model.

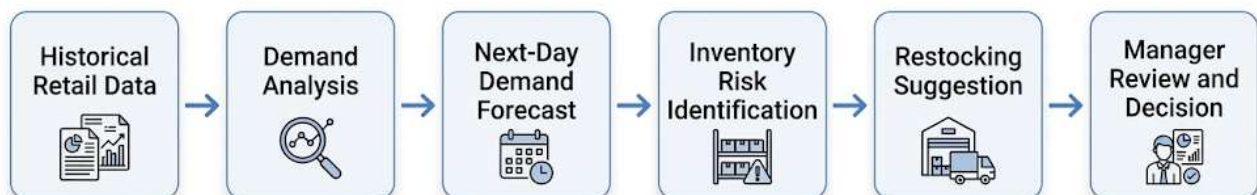
2.4 Innovation Component

The main innovation of SmartStock AI is the integration of several analytics and application components into one decision-support workflow.

The system connects historical data analysis with machine learning-based demand forecasting, inventory risk classification, restocking suggestions, business dashboards, and an AI-based explanation layer.

The Random Forest model performs the next-day demand forecasting, while the Gemini API is used to explain existing forecast, risk, and restocking results in simple language.

This creates a practical workflow:





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Project Objectives

3.1 Objectives

The main objective of SmartStock AI is to develop a data-driven inventory decision-support system that uses historical retail data to forecast demand, identify inventory risk, and provide restocking suggestions.

The specific objectives of the project are:

- To analyze historical retail sales and inventory-related data.
- To identify demand and stockout patterns across stores and products.
- To prepare and validate data for analytical and machine learning use.
- To develop features that capture historical demand, time patterns, stockout behaviour, and contextual information.
- To build a machine learning model for next-day demand forecasting.
- To identify inventory risk using forecasted demand and historical stockout patterns.
- To generate restocking suggestions using a seven-day demand cover rule.
- To store and manage project data and results using PostgreSQL.
- To develop Power BI dashboards for business analysis and reporting.
- To develop an interactive Streamlit application for inventory analysis.
- To use the Gemini API to explain existing forecast, risk, and restocking results in simple language.
- To deploy the application and database using cloud services.

3.2 Expected Outcomes and Deliverables

The expected outcomes of the project were:

- A validated and processed retail dataset.
- A machine learning feature dataset for demand forecasting.
- A trained Random Forest demand forecasting model.
- Next-day demand forecast results.
- Inventory risk classification into Low, Medium, and High levels.
- Suggested restocking quantities based on predicted demand.
- PostgreSQL database for storing project information and results.
- Power BI dashboards for business reporting.
- Streamlit-based inventory decision-support application.
- Gemini-powered AI Inventory Assistant.
- Deployed application using Streamlit Community Cloud and Neon PostgreSQL.
- GitHub repository containing the project source code.



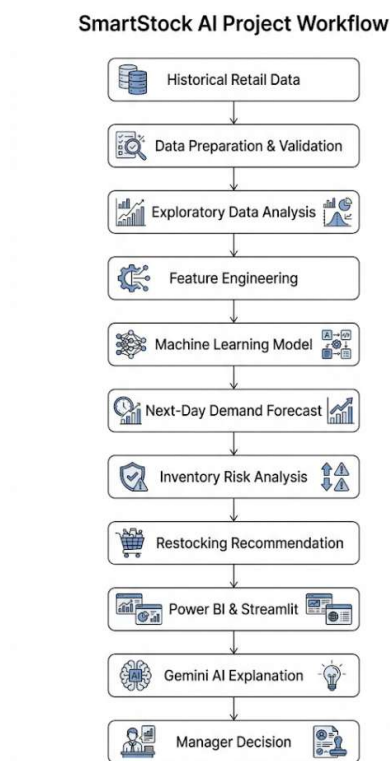
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Methodology and Results

4.1 Project Workflow

SmartStock AI follows an end-to-end data analytics workflow that converts historical retail data into inventory decision-support information.

SmartStock AI Project Workflow



The workflow begins with historical retail data and continues through data preparation and analysis. Features are then created for machine learning, and a Random Forest Regressor is used to forecast next-day demand. The forecast and historical stockout patterns are used to identify inventory risk and generate restocking suggestions.

The resulting information is presented through Power BI dashboards and the Streamlit application. The Gemini AI Inventory Assistant provides simple explanations of the existing results. The final decision remains with the manager.



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4.2 Dataset Description

The project uses two datasets for different analytical purposes.

FreshRetailNet-50K

FreshRetailNet-50K is the primary dataset used for sales analysis, stockout analysis, demand forecasting, inventory risk analysis, and restocking recommendations.

The dataset used in the project contains:

Description	Value
Training records	4,500,000
Store-product pairs	50,000
Stores	898
Products	865
Period	90 days
Date range	28 March 2024 – 25 June 2024

The dataset includes information related to sales, stock status, discounts, holidays, and weather/context information.

Instacart Market Basket Analysis

The Instacart dataset was used as a supporting dataset for analysing purchase and reorder behaviour.

It contains:

Description	Value
Orders	3.4 million
Shoppers	206,209
Products	49,688
Aisles	134
Departments	21

The Instacart and FreshRetailNet-50K datasets represent different retail environments. They were used separately and were **not directly joined**.



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4.3 Data Preparation and Data Quality

The FreshRetailNet-50K dataset was prepared and validated before using it for analysis and machine learning. The purpose of data preparation was to ensure that the data was complete, consistent, and suitable for further processing.

The validation confirmed the following:

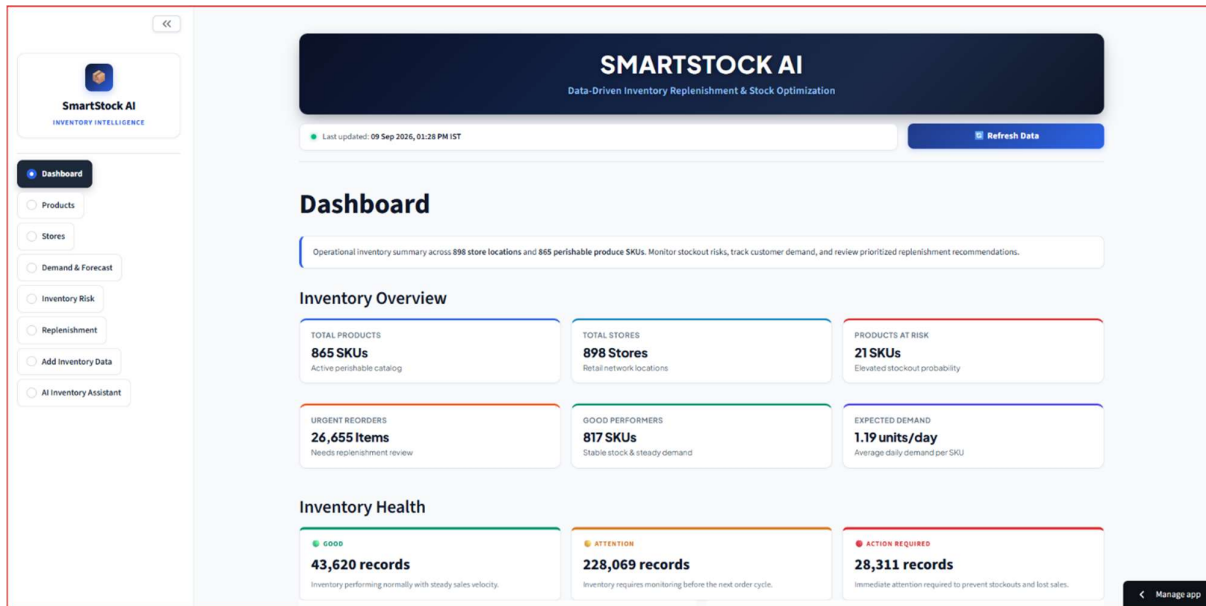
Data Quality Check	Result
Fact rows	4,500,000
Missing values	0
Duplicate business key groups	0
Negative sales	0
Zero sales	200,760
Positive sales	4,299,240
Invalid stockout-hour rows	0
Inconsistent stockout flags	0
Invalid discount rows	0
Invalid holiday/activity flags	0
Orphan stores	0
Orphan products	0
Stores	898
Products	865
Date range	28 March 2024 – 25 June 2024

The data contained no missing values, duplicate business key groups, negative sales, or invalid stockout, discount, holiday, or activity records. There were 200,760 zero-sales observations, while 4,299,240 records contained positive sales.

The validation also confirmed that all store and product references were valid, with no orphan stores or products.



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4.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the main patterns in the retail data before developing the forecasting model.

The analysis focused on:

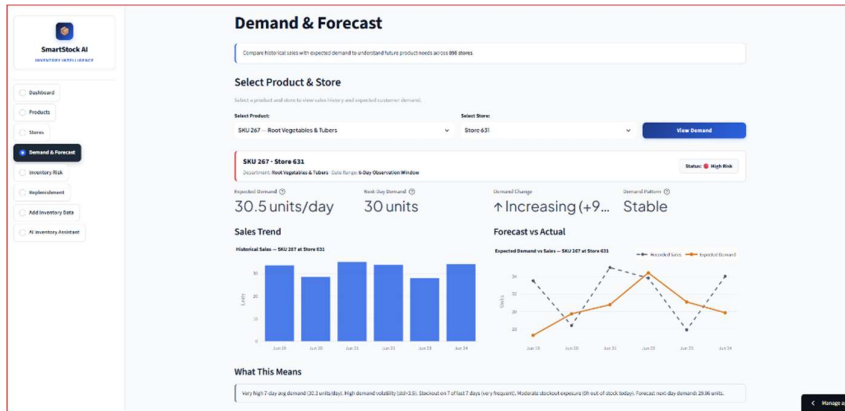
- Sales and demand patterns.
- Store-level sales behaviour.
- Product-level demand.
- Stockout patterns.
- Effects of discounts and holidays.
- Time-based demand patterns.
- Supporting purchase and reorder behaviour using the Instacart dataset.

EDA helped identify patterns in historical demand and provided a foundation for selecting relevant features for the forecasting model.

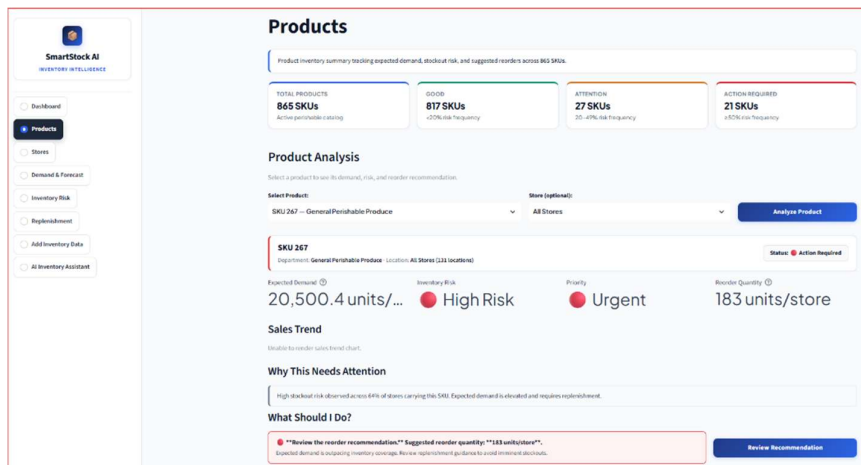


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Important Sales/Demand Trend from the Actual Project



Stockout Analysis or Store/Product Analysis from the Actual Project



4.5 Feature Engineering

Feature engineering was performed to convert the historical retail data into meaningful inputs for the demand forecasting model. The features were designed to capture recent demand patterns, time-based behaviour, stockout information, and contextual factors.

The main features used were:

- **Time Features:** Day of week, week of year, month, and weekend indicator.
- **Lag Features:** Previous demand or sales values using Lag 1, Lag 3, and Lag 7.
- **Rolling Features:** Rolling 3-day mean, rolling 7-day mean, and rolling 7-day standard deviation.
- **Stockout Features:** Previous stockout information, rolling stock hours, and rolling stock days.
- **Weather and Context Features:** Relevant contextual information available in the dataset.

The target variable for the machine learning model was **next-day demand**.



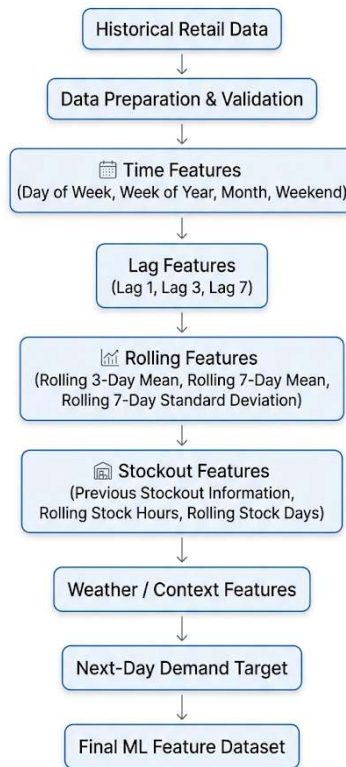
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After feature engineering, the validation contained 4.5 million rows across 50,000 store-product pairs, with approximately 4.45 million rows having a populated next-day demand target.

The final feature dataset was stored as:

`data/processed/ml_features.parquet`

Feature Engineering / Data Preparation Workflow



4.6 Machine Learning and Demand Forecasting

A **Random Forest Regressor** was used to forecast next-day demand for each store-product combination.

The model was configured with:

- **Model:** Random Forest Regressor
- **Number of trees:** 100
- **Maximum depth:** 16
- **Target:** Next-day demand

The model was trained using the engineered historical features. Its performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R^2 , and Mean Absolute Percentage Error (MAPE).



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Evaluation Metric	Result
MAE	0.3889
RMSE	0.6451
R ²	0.8814
MAPE	53.90%

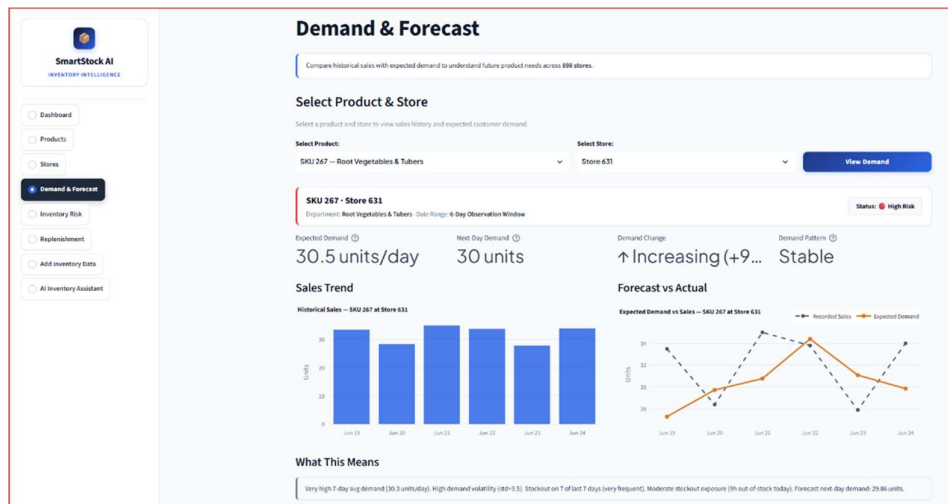
The model achieved an R² value of 0.8814, indicating that it explained a substantial proportion of the variation in the target within the evaluated data. R² is **not treated as model accuracy**.

The MAPE value is relatively high because the dataset contains a considerable number of low or zero-sales observations, where percentage-based error can become large.

The trained model was saved as:

`models/random_forest_model.pkl`

Feature importance analysis showed that the **rolling 7-day mean** was the dominant feature. Sales history, lag features, and calendar/context features also contributed to the model's predictions.





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4.7 Inventory Risk Analysis

Inventory risk analysis was performed by combining the demand forecasts with historical stockout patterns. The purpose was to identify products that may require greater attention from the manager. The system classifies inventory risk into three levels:

- **LOW**
- **MEDIUM**
- **HIGH**

The final risk distribution was:

Risk Level	Number of Records
HIGH	28,311
MEDIUM	228,069
LOW	43,620

These risk levels provide a simple way for managers to identify products that may need attention and review their associated demand and inventory information.

4.8 Restocking Recommendation

The Restocking page provides a suggested quantity based on predicted daily demand. The project uses a simple **seven-day demand cover rule**.

The calculation is:

$$\text{Suggested Restocking Quantity} = \text{Predicted Daily Demand} \times 7$$

For example, if the predicted daily demand for a product is 10 units:

$$10 \times 7 = 70 \text{ units}$$

Therefore, the suggested restocking quantity would be **70 units** for seven days of predicted demand.

The recommendation is intended to support inventory decision-making. It is **not an automatically calculated optimal purchase order** and does not consider factors such as supplier lead time, safety stock, or real-time warehouse inventory.

The manager reviews the recommendation and makes the final decision.

SmartStock AI
Inventory Replenishment

- Dashboard
- Products
- Stores
- Demand & Forecast
- Inventory Risk
- Replenishment**
- Add Inventory Data
- AI Inventory Assistant

Replenishment

Review reorder recommendations and decide what to do.

Filter Recommendations

Store Location: All Stores | Priority: Monitor | Risk Level: Medium Risk | Department: All Departments

Filter by SKU ID (optional): 267

Reset Filters

RECOMMENDATIONS
53
Matching Items

URGENT INCREASES
0
Review required

SUMMER STOCK VOLUME
9,111
Total recommended units

RECORDS TODAY
0
Saved to database

Reorder Recommendations

Review calculated replenishment recommendations, expected demand, and recorded decisions.

Product	Store	Expected Demand	Risk	Priority	Suggested Reorder	Decision Status	Reason
SKU 267 - General Perishable Produce	Store 547	31.1 unit/day	Medium Risk	Monitor	218 units	Pending Review	Very high 7-day avg demand (31.7 unit/day). High demand volatility
SKU 267 - General Perishable Produce	Store 668	29.2 unit/day	Medium Risk	Monitor	205 units	Pending Review	Very high 7-day avg demand (30.1 unit/day). High demand volatility
SKU 267 - General Perishable Produce	Store 499	29.3 unit/day	Medium Risk	Monitor	205 units	Pending Review	Very high 7-day avg demand (28.3 unit/day). High demand volatility
SKU 267 - General Perishable Produce	Store 732	29.2 unit/day	Medium Risk	Monitor	205 units	Pending Review	Very high 7-day avg demand (30.0 unit/day). High demand volatility
SKU 267 - General Perishable Produce	Store 670	28.9 unit/day	Medium Risk	Monitor	202 units	Pending Review	Very high 7-day avg demand (29.9 unit/day). High demand volatility
SKU 267 - General Perishable Produce	Store 354	28.7 unit/day	Medium Risk	Monitor	201 units	Pending Review	Very high 7-day avg demand (29.6 unit/day). High demand volatility
SKU 267 - General Perishable Produce	Store 493	28.3 unit/day	Medium Risk	Monitor	198 units	Pending Review	Very high 7-day avg demand (29.5 unit/day). High demand volatility
SKU 267 - General Perishable Produce	Store 558	28.1 unit/day	Medium Risk	Monitor	197 units	Pending Review	Very high 7-day avg demand (29.2 unit/day). High demand volatility



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4.9 Power BI Dashboard

Power BI was used as the business intelligence and reporting layer of SmartStock AI. The dashboard presents the project results in an interactive and business-focused format.

The Power BI dashboard covers:

- Key inventory and sales KPIs.
- Sales and demand trends.
- Stockout analysis.
- Store-level analysis.
- Product-level analysis.
- Category-level analysis.
- Restocking insights.

The dashboard helps users understand the overall inventory situation and explore important patterns without having to work directly with the underlying data.

4.10 Streamlit Application

A Streamlit application was developed to provide an interactive interface for SmartStock AI. The application brings together the forecasting, risk analysis, restocking recommendations, inventory data entry, and AI explanation features in one place.

The application contains eight main pages:

1. **Dashboard** – Provides an overall view of the inventory situation and important business information.
2. **Products** – Allows users to examine individual products, including demand, sales, and risk.
3. **Stores** – Provides a store-level view and allows comparison of demand and inventory risk across stores.
4. **Demand & Forecast** – Displays historical demand and next-day demand forecasts.
5. **Inventory Risk** – Highlights products according to Low, Medium, and High risk levels.
6. **Restocking** – Provides suggested restocking quantities based on predicted demand and the seven-day cover rule.
7. **Add Inventory Data** – Allows users to enter new inventory-related information. The information is validated and stored in PostgreSQL.
8. **AI Inventory Assistant** – Explains existing forecast, risk, and restocking results in simple language.

4.11 Gemini AI Inventory Assistant

The AI Inventory Assistant uses the **Gemini API** as an explanation layer for the project.

It can explain existing project results such as:

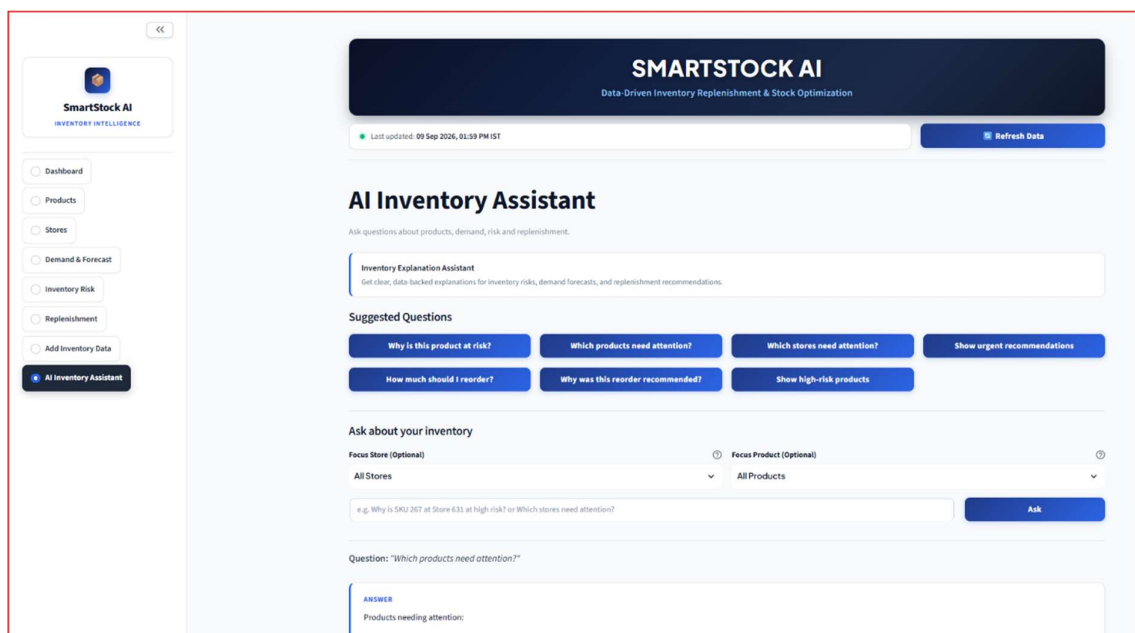
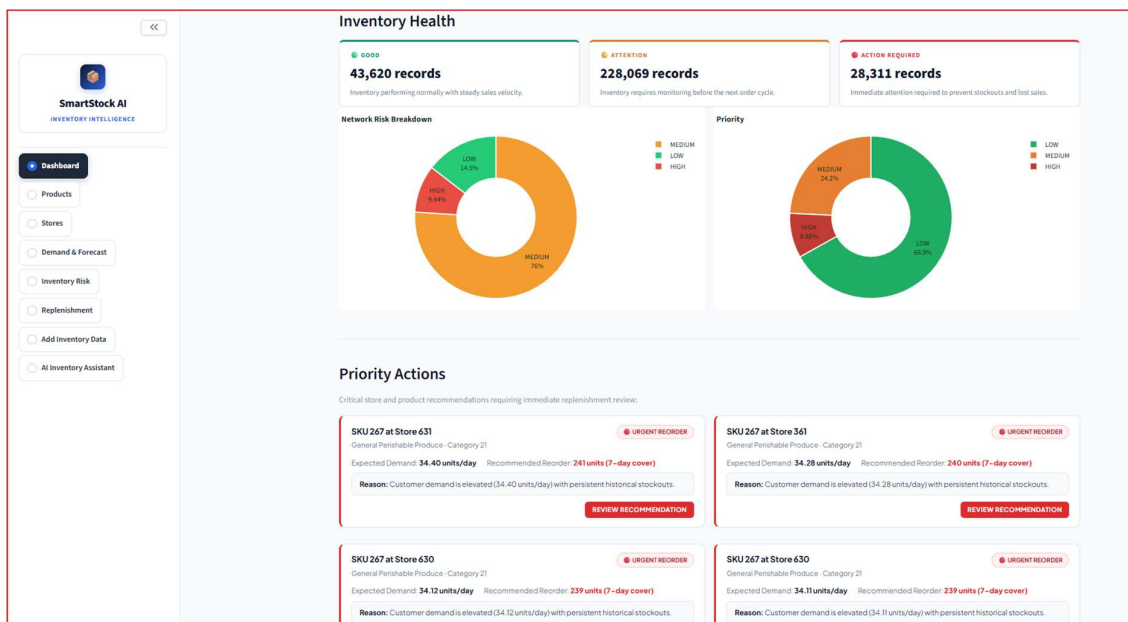
- Why a product is at risk.
- Expected demand.
- Suggested restocking quantity.
- Existing forecast and inventory results.



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Gemini does not create the main demand forecast. The **Random Forest Regressor** performs the demand forecasting, while Gemini helps communicate the existing results in simple natural language.

This separation keeps the forecasting process based on the trained machine learning model while using AI to make the results easier to understand.





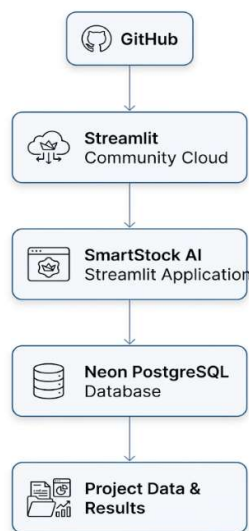
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4.12 System Architecture

SmartStock AI combines data processing, machine learning, database management, business intelligence, web application development, and AI-based explanation into a single system.

The main architecture is:

'SmartStock AI Deployment Architecture'



The analytical workflow operates alongside the application. Historical retail data is processed and used for exploratory analysis and feature engineering. The engineered features are provided to the Random Forest model to generate next-day demand forecasts. These forecasts, together with historical stockout patterns, are used for inventory risk analysis and restocking suggestions.

Power BI is used as the business intelligence and reporting layer. The Streamlit application provides an interactive interface for users, while the Gemini API provides natural-language explanations of existing project results.

4.13 Final Project Screenshots

The final report should include selected screenshots from the completed SmartStock AI application and dashboard to demonstrate the working project.

- **Dashboard:** Overall inventory and business overview.
- **Demand & Forecast:** Historical demand and next-day forecast.
- **Inventory Risk:** Low, Medium, and High risk results.
- **Restocking:** Suggested restocking quantity.
- **AI Inventory Assistant:** Gemini-based explanation of project results.
- **Power BI Dashboard:** Business intelligence and reporting view.



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4.14 GitHub Repository

The complete project source code and supporting project files are maintained in GitHub.

Personal GitHub Repository:

<https://github.com/DeepakKumar29th/SmartStock-AI>

Sure Trust Repository:

<https://github.com/sure-trust/DEEPAK-KUMAR-S-g1-data-analytics>

SmartStock AI. Deployed Streamlit Application:

<https://smartstock-inventory-ai.streamlit.app/>

The project uses environment variables and Streamlit Secrets for credentials and API keys. Sensitive credentials are not stored directly in the GitHub repository.





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Social / Industry relevance of the project

5.1 Industry Relevance

Inventory management is an important activity in the retail industry because businesses need to understand demand and maintain appropriate stock levels. SmartStock AI demonstrates how historical retail data can be used to support inventory-related decisions.

The project can help demonstrate the practical use of data analytics and machine learning for:

- Understanding sales and demand patterns.
- Identifying products with higher inventory risk.
- Forecasting expected demand.
- Supporting restocking decisions.
- Presenting business information through interactive dashboards.

5.2 Business Relevance

SmartStock AI brings different analytical outputs into one system. Instead of looking at historical sales, forecasts, and risk information separately, a manager can review these results together.

The system provides:

Demand Forecast → Inventory Risk → Restocking Suggestion → Manager Decision

This can help managers review inventory information in a more structured way.

5.3 Decision-Support Value

The project is designed as a **decision-support system**. It does not automatically purchase products or place orders.

The final decision remains with the manager, who can review the forecast, risk level, and suggested restocking quantity before taking action.

This approach keeps the system focused on providing useful information while allowing human judgment to remain part of the inventory decision-making process.

5.4 Social Relevance

The project also demonstrates how data analytics skills can be applied to practical business problems. By combining data analysis, machine learning, visualization, database management, and application development, SmartStock AI shows how technology can be used to make complex business information easier to understand and use.



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Learning and Reflection

6.1 Technical Learning

During the six-month Data Analytics Internship, I developed practical skills in **SQL, Python, Excel, Power BI, data analysis, data visualization, statistics, forecasting, predictive analytics, and machine learning**. I worked on case-based projects that helped me apply these skills to practical business problems.

I also gained experience in data cleaning, EDA, customer and sales analysis, business reporting, and dashboard development. SmartStock AI helped me combine these skills into an end-to-end project using Python, SQL, PostgreSQL, machine learning, Power BI, Streamlit, and AI.

6.2 Analytical and Problem-Solving Learning

I learned to approach analytics problems systematically by understanding the problem, checking the data, identifying patterns, selecting suitable methods, and validating results.

The internship also improved my ability to debug problems, try different approaches, and use feedback to improve my work.

6.3 LST and PAWR Learning

The **LST and PAWR sessions** helped me develop professional and workplace skills through sessions conducted by industry professionals and mentors.

Key areas included:

- Problem-solving and structured thinking
- Communication and presentation
- Workplace professionalism and accountability
- Teamwork and collaboration
- Personal branding and networking
- Interview readiness
- Negotiation and stakeholder management
- AI for workplace productivity
- AI-assisted problem-solving
- Emotional intelligence and workplace readiness

These sessions improved my confidence, communication, teamwork, and understanding of professional expectations.



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6.4 Mentorship and Community Experience

During the initial phase of my internship, I was mentored by Bhargavesh Dakka , GenAI Consultant at Deloitte USI. During the later phase, I was mentored by Nithyasri Kannathal EL , Senior Product Engineer at EsperGroup. Their feedback helped me improve my technical understanding and problem-solving approach.

As part of the internship, I also participated in community service by planting 8 long-life tree saplings and serving 8 senior citizens. These activities helped strengthen my sense of responsibility, communication, and social contribution.

6.5 Overall Internship Experience

The internship gave me practical exposure to the complete data analytics process, from data preparation and analysis to visualization, machine learning, AI, and project development.

Overall, the six-month experience improved my technical skills, problem-solving ability, communication, and confidence in working on analytics projects independently.



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Conclusion and Future Scope

7.1 Conclusion

SmartStock AI was developed as a data-driven inventory decision-support system using historical retail data. The project combines data analysis, machine learning, database management, business intelligence, web application development, and AI-based explanation in a single workflow.

The system analyzes historical demand and stockout patterns, predicts next-day demand using a Random Forest Regressor, identifies inventory risk, and provides restocking suggestions using a seven-day demand cover rule. The results are presented through Power BI and the Streamlit application, while the Gemini AI Inventory Assistant helps explain existing results in simple language.

The project successfully brings together the major skills developed during the internship and demonstrates how data can be transformed into useful information for inventory-related decision-making. The final decision remains with the manager, making the system a practical decision-support tool rather than an autonomous purchasing system.

7.2 Future Scope

The project can be further extended in the future by:

- Integrating real-time inventory or ERP data.
- Adding supplier lead-time information.
- Including safety-stock calculations.
- Developing more advanced inventory optimization methods.
- Supporting automatic model retraining when new data becomes available.
- Improving demand forecasting with additional forecasting techniques.
- Extending the system for larger and more varied retail environments.

These improvements could make the system more suitable for real-world inventory management while maintaining its role as a decision-support system.